

.NET

Interview Questions

Q: What is .NET?



A: .NET is a free, **cross-platform**, **open-source** developer platform from Microsoft used to build different types of applications, such as web, mobile, desktop, and cloud.

Q: What is the difference between .NET Framework, .NET Core, and .NET 5/6 /7+?



A: .NET Framework is **windows-only**, .NET Core is **cross-platform** and modular, and .NET 5+ is a unified platform combining the best of both.

Q: What is the CLR?



A: The Common Language Runtime (CLR) is the virtual machine **component** of .NET that manages memory, executes code, handles exceptions, and provides **type** safety.

Q: What is the CTS?



A: The Common **Type** System (CTS) defines how types are declared, used, and managed in the runtime to ensure **type** safety between different .NET languages.

Q: What is the CLS?



A: The Common Language Specification (CLS) is a set of base rules to ensure interoperability among .NET languages.

Q: What are value types and reference types in .NET?



A: Value types store data directly and are stored in the stack, while reference types store references to data and are stored on the heap.

Q: What is a namespace in C#?



A: A namespace is a container that holds classes, structs, enums, delegates, and interfaces to organize code and avoid naming conflicts.

Q: What is the difference between '==' and 'Equals()' in C#?



A: '==' compares `object` references by default, unless overloaded. 'Equals()' is a method that compares `object` values, and can be overridden.

Q: What is boxing and unboxing in .NET?



A: Boxing is converting a value **type** to an **object**; unboxing is converting an **object** back to a value **type**.

Q: What is the difference between 'const' and 'readonly'?



A: 'const' values are assigned at **compile-time** and are static by default, while 'readonly' can be assigned at runtime, usually in the **constructor**.

Q: What is the purpose of 'using' statement in C#?



A: The 'using' statement ensures proper disposal of IDisposable objects to free up unmanaged resources.

Q: What is garbage collection in .NET?



A: Garbage collection automatically manages memory by cleaning up unused objects in the heap.

Q: What are assemblies in .NET?



A: Assemblies are compiled code libraries (.dll or .exe) that contain metadata about types and resources.

Q: What is the Global Assembly Cache (GAC)?



A: The GAC is a **machine-wide** cache used to store assemblies that are intended to be shared across several applications on the system.

Q: What is an `interface` in C#?



A: An `interface` defines a contract with method and property signatures, but no implementation.

Q: What is the difference between an abstract `class` and an `interface`?



A: An abstract `class` can contain both implementation and abstraction, while an `interface` can only contain method signatures (until `C# 8` where default implementations are allowed).

Q: What are access modifiers in C#?



A: Access modifiers control the visibility of `class` members. Common ones are `public`, `private`, `protected`, `internal`, and `protected internal`.

Q: What is a delegate in C#?



A: A delegate is a **type-safe function** pointer that refers to methods with a specific signature.

Q: What is an event in C#?



A: An event is a message sent by an `object` to signal the occurrence of an action, often used with delegates.

Q: What is LINQ?



A: Language Integrated **Query** (LINQ) is a feature that allows querying collections in a **SQL-like** syntax using **C#**.

Q: What is `async/await` in C#?



A: The `async` and `await` keywords simplify asynchronous programming by allowing code to be `non-blocking` while maintaining a sequential style.

Q: What is the difference between Task and Thread?



A: A Task is a **higher-level** abstraction for asynchronous work, while Thread represents a **lower-level** system thread.

Q: What is dependency injection in .NET?



A: Dependency Injection is a design pattern used to achieve Inversion of Control (IoC), where dependencies are injected into a `class` rather than created within it.

Q: What is middleware in ASP.NET Core?



A: Middleware is software that is assembled into an application pipeline to handle requests and responses in ASP.NET Core.



25%

You're 25% through! Keep going!
Success is built one step at a time.

Q: What is the difference between .NET Core and .NET Standard?



A: .NET Core is a runtime and framework, while .NET Standard is a specification that defines a set of APIs available across all .NET implementations.

Q: What are generics in C#?



A: Generics allow you to define **type-safe** data structures without committing to actual data types until the code is instantiated in the runtime.

Q: What is the difference between IEnumerable and IQueryable?



A: IEnumerable executes queries **in-memory** and is suitable for **in-memory** collections. IQueryable executes queries against a data source and supports deferred execution.

Q: What is the difference between `var`, `dynamic`, and `object` in C#?



A: `var` is statically typed and resolved at `compile-time`, `dynamic` is resolved at runtime, and `object` is the base `class` of all types requiring casting for specific operations.

Q: What is the purpose of `async` streams in `C#`?



A: `Async` streams (using `IAsyncEnumerable<T>`) allow asynchronous iteration over a stream of data using `await foreach`.

Q: What is the difference between `string` and `StringBuilder` in C#?



A: `string` is immutable, so each modification creates a `new` instance. `StringBuilder` is mutable and more efficient for repeated modifications.

Q: What is reflection in .NET?



A: Reflection is the ability of a program to inspect and modify its own structure and behavior at runtime.

Q: What is the difference between early binding and late binding?



A: Early binding happens at `compile-time`, while late binding occurs at runtime using reflection or the `dynamic` keyword.

Q: What is the IDisposable `interface` used for?



A: IDisposable is used to release unmanaged resources explicitly via the Dispose method.

Q: How does the 'yield' keyword work in C#?



A: The 'yield' keyword returns each element one at a time in a custom iterator method without creating an entire collection.

Q: What is the difference between `==` and `ReferenceEquals` in C#?



A: '`==`' can be overloaded to compare values, while `ReferenceEquals` checks if two references point to the exact same **object** in memory.

Q: What are attributes in .NET?



A: Attributes are metadata annotations that provide additional information about code elements like classes or methods, used during reflection or by frameworks.

Q: What is dependency injection `middleware` in `ASP.NET Core`?



A: It's part of the `built-in` DI system in `ASP.NET Core` that injects dependencies into classes through `constructor` or method injection at runtime.

Q: What is the difference between transient, scoped, and singleton services in ASP.NET Core?



A: Transient: **new** instance every time. Scoped: one per **request**. Singleton: one for the lifetime of the app.

Q: What are filters in ASP.NET Core?



A: Filters are **components** that run before or after specific stages in the **request** pipeline, such as authorization, action execution, or result generation.

Q: What is Model Binding in ASP.NET Core?



A: Model Binding maps data from HTTP requests (route, query string, form, etc.) to action method parameters or model properties.

Q: What is the difference between TempData, ViewData, and ViewBag in ASP.NET MVC?



A: TempData persists for a single **request** or redirect. ViewData is a dictionary accessible during the current **request**. ViewBag is a dynamic wrapper over ViewData.

Q: What is Kestrel?



A: Kestrel is the **cross-platform** web server used by **ASP.NET** Core to handle HTTP requests directly without relying on IIS or other external servers.

Q: What is SignalR?



A: SignalR is a library for **ASP.NET** that enables **real-time** web functionality using WebSockets or fallback protocols.

Q: What is Entity Framework Core?



A: EF Core is an **open-source ORM** that allows developers to work with databases using .NET objects, eliminating the need for most SQL code.

Q: What is the difference between eager loading and lazy loading in EF Core?



A: Eager loading fetches related data immediately with the main [query](#). Lazy loading fetches it only when accessed, usually via proxies.

Q: How do you create a `migration` in EF Core?



A: You use the `dotnet ef migrations add <MigrationName>` command to scaffold a `new migration` based on `model` changes.

Q: What is the difference between AddSingleton, AddScoped, and AddTransient in dependency injection?



A: AddSingleton: one instance for app lifetime. AddScoped: one per [request](#). AddTransient: [new](#) instance every time it's requested.

Q: What is the purpose of `appsettings.json` in ASP.NET Core?



A: `appsettings.json` stores configuration data such as connection strings, settings, and environment variables in a structured JSON format.

Q: What is the purpose of Middleware Order in ASP.NET Core?



A: The order determines how requests and responses flow through the pipeline. Earlier middleware can short-circuit requests or modify responses.



50%

Halfway there! Every expert was once a
beginner.

Q: What are tag helpers in ASP.NET Core?



A: Tag Helpers enable server-side code to participate in creating and rendering HTML elements in Razor views.

Q: What is the difference between ConfigureServices and Configure in Startup.cs?



A: ConfigureServices is used to register services with the dependency injection container. Configure defines the `middleware` pipeline that handles HTTP requests.

Q: What is the Host in ASP.NET Core?



A: The Host is an **object** that encapsulates app resources like DI container, configuration, logging, and **middleware** pipeline. It controls the app lifecycle.

Q: What is IHttpContextAccessor used for?



A: IHttpContextAccessor allows access to the current HttpContext in classes where dependency injection of HttpContext is not directly available.

Q: How do you implement caching in ASP.NET Core?



A: Caching can be implemented using in-memory caching (IMemoryCache), distributed caching (IDistributedCache), or response caching middleware.

Q: What is Policy-based Authorization?



A: Policy-based authorization allows complex rules to be defined using policies containing requirements, evaluated by handlers.

Q: What is a Hosted Service in ASP.NET Core?



A: A Hosted Service is a background task that runs in the background as long as the application is running, implementing `IHostedService` or `BackgroundService`.

Q: What is a CancellationToken and how is it used?



A: CancellationToken is used to propagate cancellation requests to `async` operations or background tasks to stop execution gracefully.

Q: What is the difference between synchronous and asynchronous code?



A: Synchronous code blocks execution until it finishes, while asynchronous code allows the application to continue processing other tasks using `await`.

Q: What is the use of ConfigureAwait(false)?



A: ConfigureAwait(false) avoids capturing the synchronization context, improving performance and preventing deadlocks in certain scenarios.

Q: What are Nullable Reference Types in C#?



A: Nullable Reference Types allow you to explicitly indicate whether a reference variable may be `null`, improving `null` safety during `compile-time` checks.

Q: What is the difference between record and `class` in C#?



A: Records are reference types like classes but are immutable and provide `built-in value-based` equality by default.

Q: What is Minimal API in .NET 6+?



A: Minimal APIs provide a lightweight syntax to build HTTP APIs with less boilerplate code compared to traditional controllers in ASP.NET Core.

Q: What is source generation in .NET?



A: Source generators are **compile-time** code generation tools that generate **C#** source files to improve performance or **reduce** repetitive code.

Q: How does .NET implement memory management?



A: .NET uses automatic garbage collection to manage memory. It divides heap memory into generations (0, 1, 2) to optimize collection frequency.

Q: What is Span<T> and why is it useful?



A: Span<T> provides a **memory-safe**, fast way to handle slices of arrays or memory regions without allocations, improving performance in **high-throughput** scenarios.

Q: What are value tuples and how are they different from regular tuples?



A: Value tuples are lightweight, mutable structs used to `return` multiple values from methods. Unlike `System.Tuple`, they support deconstruction and named fields.

Q: What are .NET analyzers?



A: .NET analyzers are **Roslyn-based** tools that examine your code during compilation to enforce coding standards or provide **suggestions/errors**.

Q: What is reflection emit?



A: Reflection emit allows dynamic creation of types and methods at runtime, useful for advanced metaprogramming scenarios like dynamic proxies.

Q: What is the difference between Code First and Database First in EF Core?



A: Code First generates the database from code classes. Database First generates code (models and context) from an existing database schema.

Q: What is the Repository pattern and how is it used in .NET?



A: The Repository pattern abstracts data access logic, centralizing queries and interactions with a data store behind a consistent [interface](#).

Q: What is the Unit of Work pattern?



A: The Unit of Work pattern groups multiple operations under a single transaction, ensuring either all changes succeed or none are applied.

Q: How do you handle versioning in Web APIs?



A: Web **API** versioning can be handled via URL segments (e.g., /v1/), **query** strings, or headers using the `Microsoft.AspNetCore.Mvc.Versioning` library.

Q: What is the use of IHttpConnectionFactory?

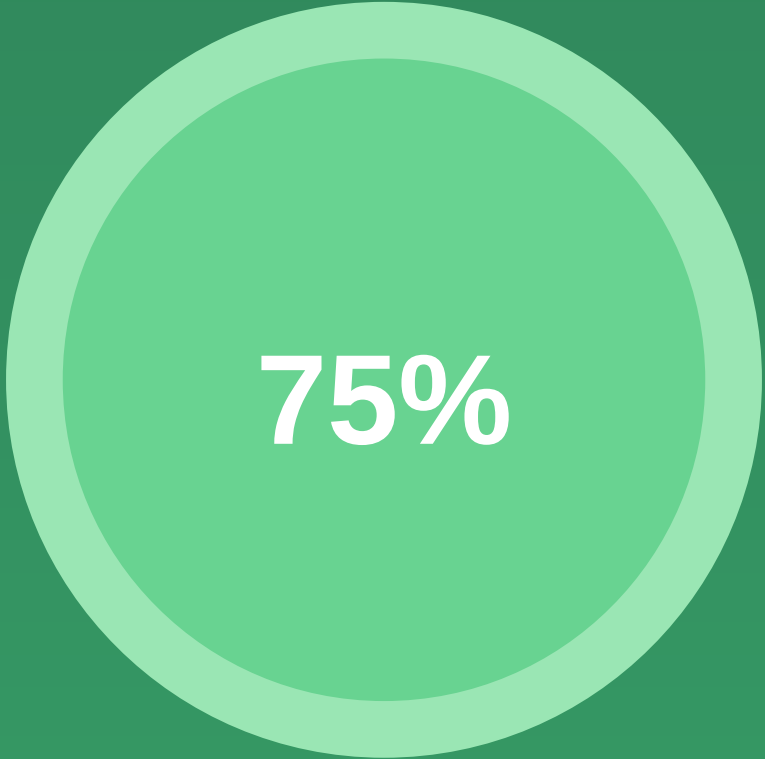


A: IHttpConnectionFactory is used to manage the lifecycle of HttpClient instances, improving performance and reliability of outbound HTTP requests.

Q: What is Blazor?



A: Blazor is a framework for building interactive web UIs using **C#** instead of **JavaScript**, supporting both **client-side** (WebAssembly) and **server-side** hosting models.



75%

You're at 75%! Almost done, push
through and finish strong!

Q: What are Razor Pages and how are they different from MVC?



A: Razor Pages are a **page-based** programming **model** in **ASP.NET** Core, simplifying the structure by combining **view** and **code-behind** into one file.

Q: How does .NET Core handle `platform-specific` dependencies?



A: .NET Core uses Runtime Identifiers (RIDs) and `runtime-specific` NuGet packages to load `platform-specific` dependencies at runtime.

Q: What is IL (Intermediate Language) in .NET?



A: IL is a **CPU-independent** set of instructions compiled from .NET languages and executed by the CLR using **Just-In-Time** (JIT) compilation.

Q: What is JIT compilation in .NET?



A: JIT (**Just-In-Time**) compilation converts IL code into native machine code at runtime for execution on the host machine.

Q: What is AOT compilation in .NET?



A: AOT (**Ahead-of-Time**) compilation converts IL into native code at build time, improving startup performance and reducing memory usage.

Q: What is the difference between Full AOT and ReadyToRun?



A: Full AOT compiles all code to native at publish time, while ReadyToRun (R2R) precompiles some code, falling back to JIT when needed.

Q: How does gRPC compare to REST in .NET?



A: gRPC is faster and uses binary protocol buffers, ideal for internal **microservices**. REST is **text-based** and better for public APIs due to wide support.

Q: How does .NET handle multithreading and concurrency?



A: .NET provides the Thread, Task Parallel Library (TPL), `async/await`, and Parallel LINQ (PLINQ) for managing concurrency and multithreading.

Q: What is a thread pool in .NET?



A: The thread pool is a pool of worker threads maintained by .NET to efficiently manage background tasks and avoid the overhead of creating **new** threads.

Q: What is lock and how does it work?



A: lock is a [C#](#) keyword that ensures that a block of code runs by only one thread at a time, using a monitor under the hood to prevent race conditions.

Q: What are deadlocks and how can they be avoided in .NET?



A: Deadlocks occur when two or more threads wait indefinitely for resources locked by each other. Avoid nested locks and use timeout or ordered locking.

Q: What is memory leak in managed code and how to detect it?



A: Memory leaks occur when references to unused objects persist, preventing GC. Tools like dotMemory and Visual Studio diagnostics help detect leaks.

Q: What is a weak reference in .NET?



A: A weak reference allows the garbage collector to collect an `object` even if a reference to it exists, avoiding memory leaks.

Q: What is Roslyn?



A: Roslyn is the .NET compiler platform that exposes APIs for code analysis, compilation, and refactoring at compile time and runtime.

Q: What is the use of analyzers and code fixes in Roslyn?



A: Analyzers inspect code during compilation and enforce rules. Code fixes suggest automated fixes or improvements to the developer.

Q: What is reflection metadata trimming in .NET?



A: Metadata trimming removes unused metadata at publish time to **reduce** application size, especially important in AOT and Blazor WebAssembly.

Q: How do you secure an ASP.NET Core application?



A: Use HTTPS, validate input, use authentication/authorization middleware, implement CSRF/XSS protection, and secure configuration/secrets with tools like Azure Key Vault.

Q: How do you integrate OAuth2 or OpenID Connect in .NET?



A: Use `middleware` like `AddAuthentication().AddJwtBearer()` or external libraries like IdentityServer, with configuration for authority and client credentials.

Q: What is IdentityServer and how is it used?



A: IdentityServer is an OpenID Connect and OAuth 2.0 framework for ASP.NET Core used to issue security tokens for authentication and authorization.

Q: How do you handle secrets in production apps?



A: Use Secret Manager during development and environment variables, Azure Key Vault, or AWS Secrets Manager in production to securely store secrets.

Q: What is distributed tracing in `microservices` with .NET?



A: Distributed tracing tracks requests across `microservices` using tools like OpenTelemetry, Application Insights, or Jaeger to monitor system performance and issues.

Q: What is rate limiting and how is it implemented in ASP.NET Core?



A: Rate limiting restricts the **number** of requests a client can make. It can be implemented using custom **middleware** or libraries like **AspNetCoreRateLimit**.

Q: What is CQRS and how is it implemented in .NET?



A: CQRS (Command **Query** Responsibility Segregation) separates read and write logic. In .NET, it's often implemented with MediatR and separate data models.

Q: What is DDD and how does .NET support it?



A: DDD (**Domain-Driven** Design) focuses on complex business logic with domain models. .NET supports DDD with aggregates, repositories, value objects, and bounded contexts.

Q: What are bounded contexts in DDD?



A: A bounded context defines a boundary around a specific domain **model** where terms and rules are consistent, allowing separation between subdomains.

Q: How do you scale ASP.NET Core applications?



A: Scale via load balancing, horizontal scaling (multiple instances), caching, database optimization, stateless services, and containerization with Kubernetes or Docker.

Thank You!

You've completed all the questions.